

AWS Lambda Release Automation — Lab Report

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Overview

This document provides a practical, interview-ready walkthrough of a release automation workflow on AWS using Lambda, IAM, CloudWatch, and Linux/Bash scripting. It reflects a Release Engineer mindset: deploy reliably, manage versions, monitor behavior, and roll back safely, with clear documentation and process discipline.

What This Project Showcases

- Automated deployments with AWS CLI and Bash (idempotent scripts).
- Versioned AWS Lambda releases with alias-based routing (`prod``).
- Release verification via `/health`` and `/version`` endpoints.
- Rollback control to a stable version with a single command.
- Operational monitoring with Amazon CloudWatch (console and CLI).
- Troubleshooting real-world blockers (auth on alias URLs, update race conditions).
- Clear documentation, repeatability, and observability.

Screenshot #1 – High-level architecture or repository homepage (optional).

Prerequisites

- AWS account with IAM permissions to create/update Lambda, Function URLs, and CloudWatch logs.
- AWS CloudShell (recommended) or local terminal with AWS CLI installed.
- IAM role for Lambda execution (e.g., `lambda-basic-role``) attached with the AWS managed policy `AWSLambdaBasicExecutionRole``.

Prerequisite Screenshot #1-(Lambda IAM role for CloudWatch)

The screenshot displays the AWS IAM console interface. The left-hand navigation pane shows the 'Identity and Access Management (IAM)' section, with 'Policies' selected. The main content area is titled 'AWSLambdaBasicExecutionRole' and includes a description: 'Provides write permissions to CloudWatch Logs.' Below this, the 'Policy details' section shows the role's type as 'AWS managed', its creation time as 'April 09, 2015, 08:03 (UTC-07:00)', and its edited time as 'April 09, 2015, 08:03 (UTC-07:00)'. A green 'ARN copied' notification is visible next to the role's ARN: 'arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole'. The 'Permissions' tab is active, showing 'Permissions defined in this policy' with a search bar and a table of permissions. The table lists 'CloudWatch Logs' as the service, 'Limited: Write' as the access level, 'All resources' as the resource, and 'None' as the request condition. The bottom of the console shows the footer with copyright information and links to CloudShell, Feedback, Privacy, Terms, and Cookie preferences.

Policy details

Type	Creation time	Edited time	ARN
AWS managed	April 09, 2015, 08:03 (UTC-07:00)	April 09, 2015, 08:03 (UTC-07:00)	arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole

Permissions defined in this policy

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it.

Search

Allow (1 of 450 services) ☐ Show remaining 449 services

Service	Access level	Resource	Request condition
CloudWatch Logs	Limited: Write	All resources	None

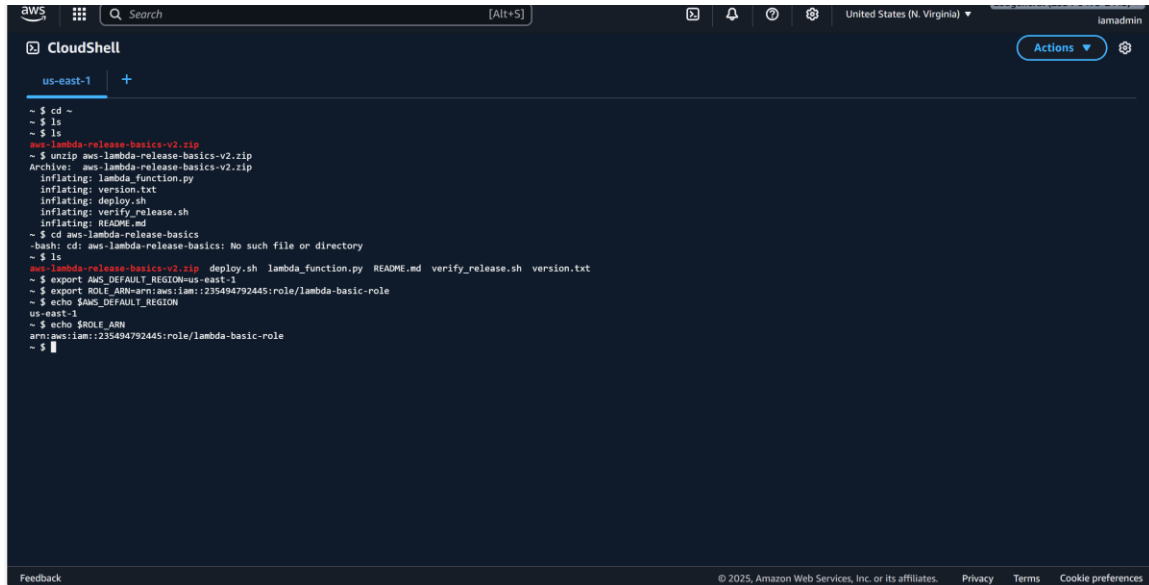
Step-by-Step Implementation

Step 0 — Setup (CloudShell + IAM)

Configure credentials, region, and role. This binds your terminal to AWS with least-privilege access and sets a consistent deployment region.

```
aws configure
export AWS_DEFAULT_REGION=us-east-1
export ROLE_ARN=arn:aws:iam::<your-account-id>:role/lambda-basic-role
echo $AWS_DEFAULT_REGION
echo $ROLE_ARN
```

Screenshot #2 – CloudShell open and environment variables exported.



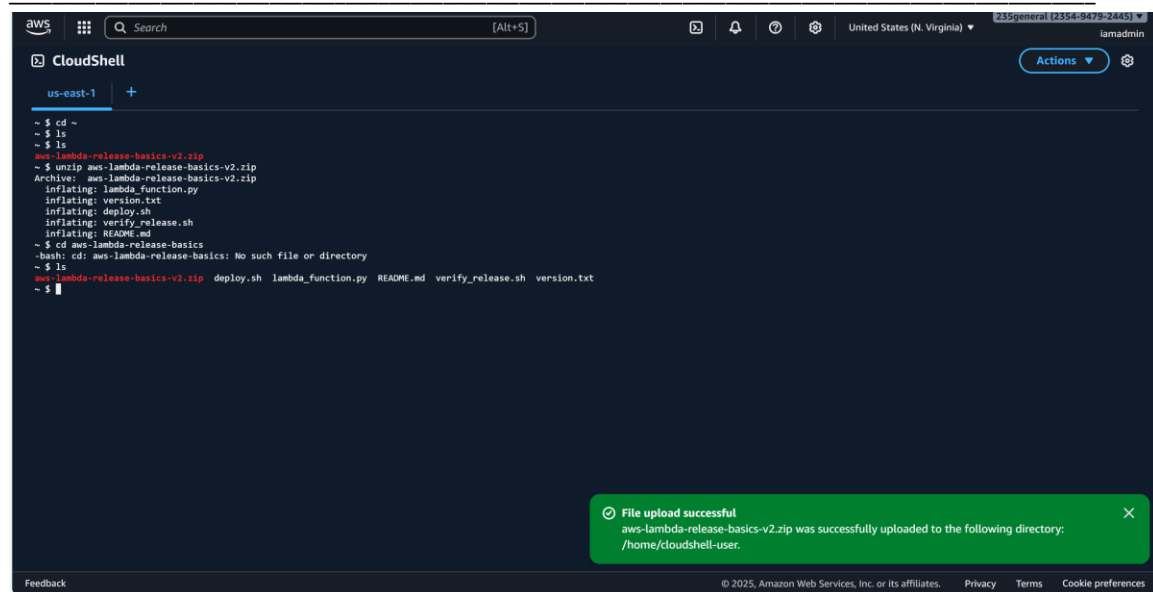
Step 1 — Upload & Unpack

Upload the project ZIP into CloudShell, extract it, and review the files.

```
unzip aws-lambda-release-basics-v2.zip
cd aws-lambda-release-basics
ls -l
```

Expected files: `lambda_function.py`, `deploy.sh`, `verify_release.sh`, `version.txt`, `README.md`

Screenshot #3 – Project files visible in CloudShell.



What's Inside the ZIP (Explained)

- `lambda_function.py` — Application code. Routes `/health` to `{'status':'ok'}`, `/version` to the version from `version.txt`, and returns a helpful JSON for other paths. Provides simple release verification endpoints.
- `version.txt` — Single source of truth for the release number (e.g., 1.0.0). The deployment script embeds this into the packaged artifact for traceability.
- `deploy.sh` — Idempotent deployment script: packages the function, creates/updates the Lambda, waits for active state, publishes an immutable version, updates alias `prod`, ensures a Function URL exists, and grants public invoke permission for the alias URL.
- `verify_release.sh` — Post-release smoke tests that call your function URL to validate `/health` and `/version`.
- `README.md` — Documentation of the workflow and commands.

Step 2 — First Release (v1.0.0)

Set the release version, deploy, publish a version, and update the `prod` alias.

```
echo "1.0.0" > version.txt
bash deploy.sh
```

Representative output:

```
[*] Packaging Lambda function...
[*] Creating new Lambda function...
[*] Waiting 30 seconds for Lambda to finish updating...
[*] Waiting for Lambda to become fully active before publishing
version...
[*] Published version 1
[*] Updated alias 'prod' -> version 1
[✓] Deploy complete.
    PROD URL: https://<your-lambda-url>/
```

Screenshot #4 – Deploy complete with PROD URL shown.

aws

Search

[Alt+S]

United States (N. Virginia)

235Generat (2354-9479-2445)

iamadmin

CloudShell

Actions

us-east-1

+

```
~ $ cd ~
~ $ ls
~ $ ls
aws-lambda-release-basics-v2.zip
~ $ unzip aws-lambda-release-basics-v2.zip
Archive:  aws-lambda-release-basics-v2.zip
  inflating: lambda_function.py
  inflating: version.txt
  inflating: deploy.sh
  inflating: verify_release.sh
  inflating: README.md
~ $ cd aws-lambda-release-basics
-bash: cd: aws-lambda-release-basics: No such file or directory
~ $ ls
aws-lambda-release-basics-v2.zip  deploy.sh  lambda_function.py  README.md  verify_release.sh  version.txt
~ $ export AWS_DEFAULT_REGION=us-east-1
~ $ export ROLE_ARN=arn:aws:iam::235494792445:role/lambda-basic-role
~ $ echo $AWS_DEFAULT_REGION
us-east-1
~ $ echo $ROLE_ARN
arn:aws:iam::235494792445:role/lambda-basic-role
~ $ echo "1.0.0" > version.txt
~ $ cat version.txt
1.0.0
~ $
```

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CloudShell

Actions

us-east-1

+

```
~ $ cd ~
~ $ ls
~ $ ls
aws-lambda-release-basics-v2.zip
~ $ unzip aws-lambda-release-basics-v2.zip
Archive:  aws-lambda-release-basics-v2.zip
  inflating: lambda_function.py
  inflating: version.txt
  inflating: deploy.sh
  inflating: verify_release.sh
  inflating: README.md
~ $ cd aws-lambda-release-basics
-bash: cd: aws-lambda-release-basics: No such file or directory
~ $ ls
aws-lambda-release-basics-v2.zip  deploy.sh  lambda_function.py  README.md  verify_release.sh  version.txt
~ $ export AWS_DEFAULT_REGION=us-east-1
~ $ export ROLE_ARN=arn:aws:iam::235494792445:role/lambda-basic-role
~ $ echo $AWS_DEFAULT_REGION
us-east-1
~ $ echo $ROLE_ARN
arn:aws:iam::235494792445:role/lambda-basic-role
~ $ echo "1.0.0" > version.txt
~ $ cat version.txt
1.0.0
~ $ bash deploy.sh
[*] Packaging Lambda function...
[*] Updating existing Lambda function...
[*] Waiting 30 seconds for Lambda to finish updating...
[*] Waiting for Lambda to become fully active before publishing version...
[*] Published version 1
[*] Updated alias 'prod' -> version 1
[✓] Deploy complete.
prod url: https://qebgunnr6r7umfttqcqgfjmw@tlpwo.lambda-url.us-east-1.on.aws/
~ $
```

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Step 3 — Release Verification

Run smoke tests to validate health and version end-to-end.

```
curl [your-lambda-url-here]/health
curl [your-lambda-url-here]/version
```

Expected:

```
{"status":"ok"}
{"version":"1.0.0"}
```

Screenshot #5 – curl output for /health and /version.

***Tips: Troubleshooting & Hardening**

Two real issues were handled and automated in the scripts:

- 1) ResourceConflictException (update in progress): AWS internally locks functions during updates. Added waits (`sleep` + `aws lambda wait function-active`) to avoid race conditions.
- 2) Function URL 'Forbidden' when using alias `prod`: Alias URLs need explicit permission. Added `aws lambda add-permission` for the alias URL with `--function-url-auth-type NONE`.

```
aws lambda add-permission --function-name release-basics-
function:prod --action lambda:InvokeFunctionUrl --principal "*" -
-function-url-auth-type NONE --statement-id allow-public-access
```

Step 4 — Second Release (v1.1.0)

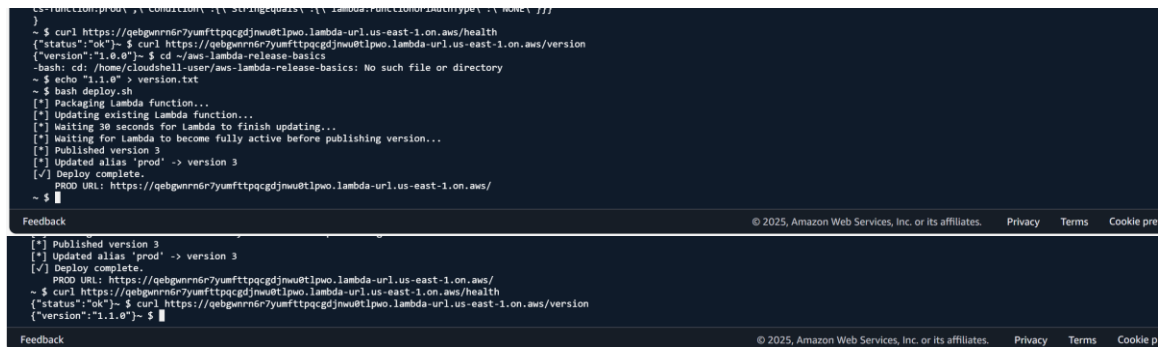
Simulate a new sprint release by bumping the version and deploying, then re-verify the `/version` endpoint.

```
echo "1.1.0" > version.txt
bash deploy.sh
curl [your-lambda-url-here]/version
```

Expected:

```
{"version": "1.1.0"}
```

Screenshot #7 – Deploy output for v1.1.0 and version check.



```
CS-Function:prod) {\ condition: {\ StringEquals: {\ Lambda::Function::Architecture: {\ Name: }}}
}
~ $ curl https://qebgmnrn6r7umfttqcgjnuw@tlpwo.lambda-url.us-east-1.on.aws/health
{"status":"ok"}~ $ curl https://qebgmnrn6r7umfttqcgjnuw@tlpwo.lambda-url.us-east-1.on.aws/version
{"version":"1.1.0"}~ $ cd ~/aws-lambda-release-basics
-bash: cd: /home/cloudshell-user/aws-lambda-release-basics: No such file or directory
~ $ echo "1.1.0" > version.txt
~ $ bash deploy.sh
~ $
[*] Packaging Lambda function...
[*] Updating existing Lambda function...
[*] Waiting 30 seconds for Lambda to finish updating...
[*] Waiting for Lambda to become fully active before publishing version...
[*] Published version 3
[*] Updated alias 'prod' -> version 3
[✓] Deploy complete.
INFO URL: https://qebgmnrn6r7umfttqcgjnuw@tlpwo.lambda-url.us-east-1.on.aws/
~ $

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[*] Published version 3
[*] Updated alias 'prod' -> version 3
[✓] Deploy complete.
INFO URL: https://qebgmnrn6r7umfttqcgjnuw@tlpwo.lambda-url.us-east-1.on.aws/
~ $ curl https://qebgmnrn6r7umfttqcgjnuw@tlpwo.lambda-url.us-east-1.on.aws/health
{"status":"ok"}~ $ curl https://qebgmnrn6r7umfttqcgjnuw@tlpwo.lambda-url.us-east-1.on.aws/version
{"version":"1.1.0"}~ $

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```

Step 5 — View Version History

List all published versions for auditability and rollback reference.

```
aws lambda list-versions-by-function --function-name release-basics-
function --query
"Versions[*].{Version:Version,Description:Description}"
```


Screenshot #8 – Version list output.

```
> --function-name release-basics-function \
> --query "Versions[0].Version,Description"
{
  "Version": "$LATEST",
  "Description": "",
},
{
  "Version": "1",
  "Description": "Release 1.0.0"
},
{
  "Version": "2",
  "Description": "Release 1.0.0"
},
{
  "Version": "3",
  "Description": "Release 1.1.0"
}
~ $
```

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Step 6 — Rollback Simulation

Repoint `prod` to a known good version for rapid recovery; confirm the version endpoint.

```
aws lambda update-alias --function-name release-basics-function --
name prod --function-version 1
```

```
curl [your-lambda-url-here]/version
```

Expected:

```
{"version":"1.0.0"}
```

Screenshot #9 – Rollback result showing version 1.0.0.

```
}
}
~ $ aws lambda update-alias \
> --function-name release-basics-function \
> --name prod \
> --function-version 1
{
  "AliasArn": "arn:aws:lambda:us-east-1:235494792445:function:release-basics-function:prod",
  "Name": "prod",
  "FunctionVersion": "1",
  "Description": "",
  "RevisionId": "446db6ff-c1f3-4854-864b-35c55f10d777"
}
~ $ curl https://qehgummr6r7umfttqcdjmwu@lpmo.lambda-url.us-east-1.on.aws/version
{"version":"1.0.0"}~ $
```

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Step 7 — CloudWatch Monitoring

Post-release, inspect logs to confirm healthy execution and spot anomalies.

Console: CloudWatch → Log groups → /aws/lambda/release-basics-function

```
aws logs filter-log-events --log-group-name "/aws/lambda/release-
basics-function" --limit 5 --query "events[*].message"
```

Screenshot #10 – CloudWatch log group and a recent log stream.

CloudWatch

Log groups

/aws/lambda/release-basics-function

CloudWatch

Log groups

/aws/lambda/release-basics-function

Log group details

Log class

Standard

ARN

arn:aws:logs:us-east-1:235494792445:log-group:/aws/lambda/release-basics-function:

Creation time

14 minutes ago

Retention

Never expire

Stored bytes

-

Metric filters

0

Subscription filters

0

Contributor Insights rules

-

KMS key ID

-

Anomaly detection

Configure

Data protection

-

Sensitive data count

-

Custom field indexes

Configure

Transformer

Configure

Log streams

Tags

Anomaly detection

Metric filters

Subscription filters

Contributor Insights

Data protection

Log streams (3)

By default, we only load the most recent log streams.

Filter log streams or try prefix search

Exact match

Show expired

Info

1

CloudWatch

Log groups

/aws/lambda/release-basics-function

2025/11/05/[3]dfc3ebdca8844b2ab4f4aab645329890

CloudWatch

Log groups

/aws/lambda/release-basics-function

2025/11/05/[3]dfc3ebdca8844b2ab4f4aab645329890

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

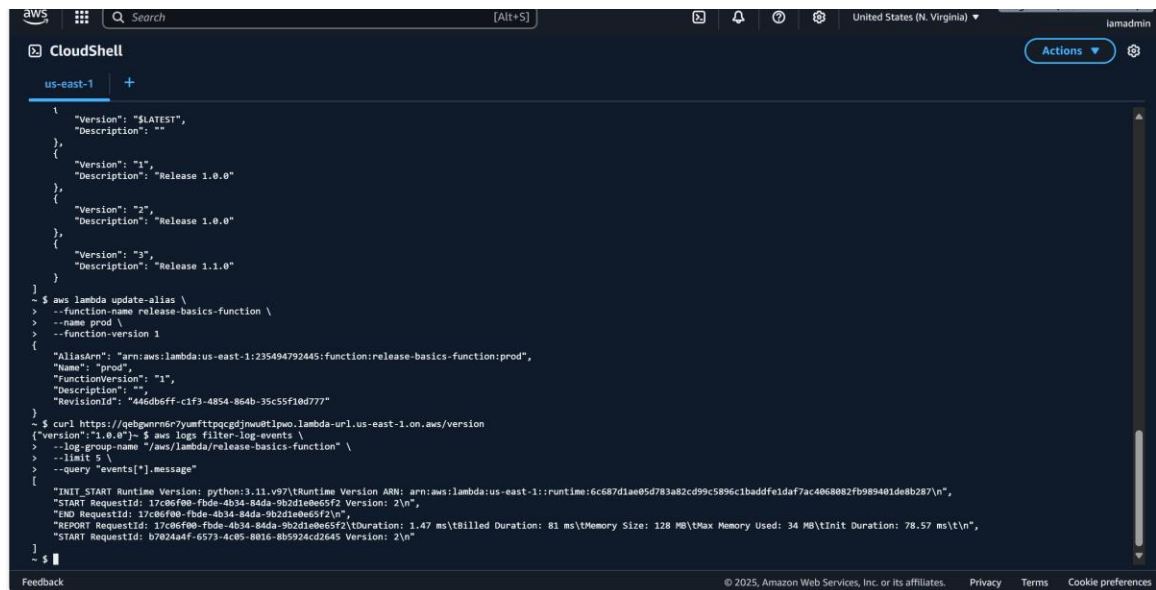
1m

1h

UTC timezone

Display

Timestamp	Message
No older events at this moment. Retry	
2025-11-05T03:12:59.756Z	INIT_START Runtime Version: python:3.11.v97 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:6c687d1ae05d783a82c...
2025-11-05T03:12:59.848Z	START RequestId: 057a582a-6bc0-4106-ad09-fbdfdcfcaf39 Version: 3
2025-11-05T03:12:59.850Z	END RequestId: 057a582a-6bc0-4106-ad09-fbdfdcfcaf39
2025-11-05T03:12:59.850Z	REPORT RequestId: 057a582a-6bc0-4106-ad09-fbdfdcfcaf39 Duration: 1.52 ms Billed Duration: 91 ms Memory Size: 128 MB Ma...
2025-11-05T03:12:59.929Z	START RequestId: 82df8e43-f6cb-4830-95dc-3d4f6f808e75 Version: 3
2025-11-05T03:12:59.931Z	END RequestId: 82df8e43-f6cb-4830-95dc-3d4f6f808e75
2025-11-05T03:12:59.931Z	REPORT RequestId: 82df8e43-f6cb-4830-95dc-3d4f6f808e75 Duration: 1.18 ms Billed Duration: 2 ms Memory Size: 128 MB Max...
No newer events at this moment. Auto retry paused. Resume	



```
aws [Alt+S] United States (N. Virginia) iamadmin
CloudShell
us-east-1 +
{
  "Version": "$LATEST",
  "Description": ""
},
{
  "Version": "1",
  "Description": "Release 1.0.0"
},
{
  "Version": "2",
  "Description": "Release 1.0.0"
},
{
  "Version": "3",
  "Description": "Release 1.1.0"
}
}

~ $ aws lambda update-alias \
> --function-name release-basics-function \
> --name prod \
> --function-version 1
{
  "AliasArn": "arn:aws:lambda:us-east-1:235494792445:function:release-basics-function:prod",
  "Name": "prod",
  "FunctionVersion": "1",
  "Description": "",
  "RevisionId": "446db6ff-c1f3-4854-864b-35c55f10d777"
}

~ $ curl https://qebgmnr6r7ymftpqcdjmwetlpw.lambda-url.us-east-1.on.aws/version
{"version":"1.0.0"}~ $ aws logs filter-log-events \
> --log-group-name "/aws/lambda/release-basics-function" \
> --limit 5 \
> --query "events[*].message"
[
  "INIT START Runtime Version: python3.11.v07\tRuntime Version ARN: arn:aws:lambda:us-east-1:runtime:6c687d1ae05d783a82cd99c5896c1baddfda7ac4068082fb989401de8b287\n",
  "START RequestId: 17c06f00-fbde-4b34-84da-9b2d1e0e65f2 Version: 2\n",
  "END RequestId: 17c06f00-fbde-4b34-84da-9b2d1e0e65f2\n",
  "REPORT RequestId: 17c06f00-fbde-4b34-84da-9b2d1e0e65f2\tDuration: 1.47 ms\tBilled Duration: 81 ms\tMemory Size: 128 MB\tMax Memory Used: 34 MB\tInit Duration: 78.57 ms\t\n",
  "START RequestId: b7034a4f-6573-4c05-8016-8b5924cd2645 Version: 2\n"
]
```

Reflective Alignment with Veeva’s Release Engineering Role

This project strengthened my release mindset: small, predictable changes; explicit waits for asynchronous AWS operations; and end-to-end verification using health checks and CloudWatch. It directly maps to Veeva's expectations for an Associate Release Engineer:

- Coordinate releases & versions → Versioned Lambda with alias `prod`, explicit publish + alias updates.
- Automate repeatable tasks → Idempotent Bash scripts with built-in waits to avoid race conditions.
- Resolve blockers quickly → Diagnosed 'Forbidden' on alias URLs and added scoped permissions; handled update locks.
- Verify and monitor → Health/version endpoints + CloudWatch logs (console and CLI).
- Document clearly → README and this guide with step-by-step and screenshot anchors.

Key takeaway: Release Engineering is about trust and control—ensuring every deployment is traceable, reversible, and observable. I treated each blocker as a signal to improve the pipeline, not a one-off fix.

Appendix — Command Reference

Most-used commands during the project:

```
aws lambda list-versions-by-function --function-name release-basics-  
function  
aws lambda update-alias --function-name release-basics-function --name  
prod --function-version <N>  
aws logs filter-log-events --log-group-name "/aws/lambda/release-  
basics-function" --limit 5 --query "events[*].message"  
aws lambda add-permission --function-name release-basics-function:prod  
--action lambda:InvokeFunctionUrl --principal "*" --function-url-auth-  
type NONE --statement-id allow-public-access
```

Tip: Keep these scripts and commands in version control and iterate as you learn.